Ali Al Shammaa

Technical Skills

Languages: Python, R , C++, SQL
ML Tools: Apache Spark, Databricks, NumPy, Pandas, Tensorflow, Keras
Developer Tools: Azure, Bash, Git, Jupyter, SSRS Report Builder, Power BI

Education

University of Waterloo | **Bachelor of Computer Science** — 2020 - 2025
Relevant Courses : Calculus III, Probability, Linear Algebra — *GPA : **91** %*

Experience

Data Scientist — May 2021 - Aug 2021
Gore Mutual Insurance — *Cambridge, ON*

- Utilised **Pandas** for ETL in Databricks followed by **SQL** for database queries to supply data into SSRS. Moreover, built pipelines in Azure with **SQL** to pull data from DBs, followed by creating visual metrics in **Power BI** with Dax.
- Delivered **6** descriptive and historical reports for performance management, auditing and operational tracking in collaboration with the analytics team, effectively quickening business operations by **200**%.
- Automated the **E2E** delivery of market insights to the managers, featuring Competitive Pricing Index, Loss Ratio and other **KPI**s in **SSRS**. Successfully, eliminated **1 week** worth of monthly raw data analysis.

Frontend Developer — Jul 2020 - Sep 2020
Atlas365 Inc. — *Mississauga, ON*

- Improved the UI design of **5** components in an Angular website that boosted user's duration of visit.

Automation Developer — May 2020 - Nov 2020
GamePill — *Markham, ON*

- Created 3 Social Media bots in **Python** with Selenium, attracting testing streamers **4** times more.

VR Developer — Aug 2019 - Sep 2019
LiTrans Lab - Ryerson University — *Toronto, ON*

- Enhanced the photorealism of a VR simulation for AVs, achieving a 3 fold improvement in the drivers' reaction accuracy.

Academic Research

ML Researcher — May 2022 - present
Inholland University — *Delft, Netherlands*

- Investigating the success of several Machine Learning techniques including **Linear Regression, SVM, Random Forrest, Time Series Analysis and Neural Networks** in predicting harvest quality.

Projects

Watonomous | UW's Autonmous Vehicle Team

- Working with the team on Object Detection, Action Recognition and Action classification in the vehicle's environment using ROAD dataset.
- Implementing novel neural architectures to accelerate autonomous software learning, hence increasing the vehicle's decisions accuracy and safety.

ChessAI | Online Chess Engine

- Implemented an online Chess engine in JavaScript using a Minimax Algorithm with AlphaBeta pruning, enhanced via Zobrist Hashing, iterative deepening framework, killermove heuristics and move reordering.

Image Recognition | Neural Nets

- Built multiple ANNs and CNNs using **Tensorflow** and **Keras**. Achieved **97** % on MNIST with ANN. Achieved **53 %** and **70 %** on CIFAR-10 with ANN and CNN respectively.

CNN First Principles | Neural Net

- Built a CNN with b2b convolutional, pooling and fully connected layers from first principles using only **NumPy** and implemented backpropagation to achieve **90**% success rate on odd MNIST.